GlucoBloom: System Architecture & Data Flow

Personalized Digital Twin for Continuous Glucose Forecasting • Unstop Digital Twin Challenge 2026

1. SENSOR INGESTION

- Continuous CGM (5m)
- Carbohydrate Logs
- Step / Exercise Rate
- Heart Rate & HRV
- Sleep Duration & Quality

Data Quality Guard:

- Dropout Detection (>15m)
- Compression Outlier Filter
- Strict Time Regularization

2. FEATURE VECTOR

- 12 Lags: t-55m to t
- 15m & 30m Velocities
- Rolling Mean & Volatility
- Circadian Diurnal sin/cos
- Cumulative 30m Carbs
- Cumulative 30m Steps

Split Protocol:

- Strict Patient-Holdout
- No Random Row Leakage

3. TWIN AI ENGINE

Level 1: General Model

- XGBoost Regressor
- 160 Train Patients
- 7.64 mg/dL MAE

Level 2: Personal Adapter

- Fine-tuned on User CGM
- 6.44 mg/dL MAE (+26.4%)

Uncertainty Quantiles:

- 10th & 90th Percentiles
- 80% Cone (85% Coverage)

4. CLINICAL UI

- 30m Forecast Cone
- Twin Evolution
- Time Machine Replay
- **Clarke Error Grid** (100% Zones A+B)
- Fan of Futures Sim
- Food & Carb Diary
- AGP Clinical Export
- Fast API + Streamlit